MAHATMAGANDHIUNIVERSITY,KOTTAYAM

MGU-UGP(HONOURS)

THIRDSEMESTEREXAMINATION

(2024ADMISIONONWARDS)

MG3DSEECT205 MicrocontrollerProgramming

Duration:1.30hours **MaximumMarks:50**

**Remember(K),Understand(U),Apply(A),Analyse(An),Evaluate(E),Create(C),Skill(S),Interest (I)andAppreciation(Ap)*

Part A

Short Answer Questions

Answer any **10** Questions

Eachquestioncarries2marks

1.	Brieflyexplainthesequencetypes.	U	1
2.	WhatareComparisonoperators,andtheirtypes.	U	1
3.	DescribethegeneralstructureofaPythonprogram.	U	1
4.	Given two integers X = 5 and Y = 3, perform the following bitwise operations: a. BitwiseAND(X&Y) b.BitwiseOR (X|Y)	A	1
5.	Convert the binary number 1111011 to decimal and the decimal number 123 to binary.	A	2
6.	WhatdoyoumeanbyGPIORegister?NamecommonlyusedGPIORegisters.	U	2
7.	Whatarethestringmethodsusedforthefollowing operations: a. Convertingastringtouppercase b. Convertingastringtolowercase c. Splittingastringintoalistofwords d. Removingleadingandtrailingwhitespacefrom a string	U	2
8.	Writeaprogramthatusesawhilelooptoprintnumbersfrom0to4.	A	2
9.	Mentiontheprimarymethodsforcustomizingwidgetproperties.	U	3
10.	Givethesyntaxforthewidgets:Label,ButtonandEntry.	U	3
11.	WhatisTkinter,andwhyisitusedforbuildingGUIapplicationsinPython?	U	3
12.	What are some common options used to customize the properties of Tkinter widgets?	U	3
13.	ListthebasicstepstocreateaTkinterPython app.	U	3
14.	NametheTkinterwidgetsusedforthefollowing purposes: a. Displayingtextorimages. b. Creatingclickable controls.	U	3

2*10 = 20

Part B

Short Essay Questions

Answer any **6** Questions

Eachquestioncarries5marks

15.	DescribethesyntaxrulesinPython.Howdotheserulescontribute to writing clean and readable code?	U	1
16.	DescribethekeyfeaturesandportlayoutofRaspberryPi4.Explain the purpose of each port and how they can be used.	U	1
17.	Explainthedifferencebetweenarithmetic,logicalandassignment operators in programming, providing examples of each type of operator.	U	1
18.	WriteaprogramthatusesdigitalReadtoreadthestateofabutton connected to digital pin 2 and digitalWrite to control an LED connected to digital pin 13.	A	2
19.	DescribethestructureofaGPIOpin,explainthedifferentmodesit can operate in, and discuss the various operations that can be performed on a GPIO pin.	U	2
20.	ExplainstringformattingandConversioninthe context of number systemapplications,includingbinary,octal,andhexadecimal.Provide a simple example for each	U	2
21.	In Tkinter, there are three geometry managers that can be used to arrangewidgetsinaGUIapplication.Explainthepurposeandusageof each geometry manager.	U	3
22.	ExplainthepurposeandusageoftheTkinter widgets,providing examples for each.	U	3

6***5 = 30**